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MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M11, ASSEMBLY

1. SCOPE

1.1 This specification covers the Thruster, Cartridge Actuated, M11, Assembly, loaded with Cartridge, Thruster, M141 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MS-28741 - Hose Assembly Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10521565 - Thruster, Cartridge Actuated, M11, Assembly

FSC 1377

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2.1 (Cont'd)

C8800969 - Container, Ammunition, Fiber M434 for Thruster Cartridge Actuated, M11 with Cartridge M141.

C8800970 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M11 with Cartridge M141 in Fiber Container M434

IEL-10521565 - Index of Inspection Equipment List for Thruster, Cartridge Actuated, M11, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Torque.- Torque requirements shall be as specified in Table I. No movement shall occur upon application of the breakaway torque.

Table I

Components	A Run-on (inch-pounds) (Minimum)	B Seating or Bottoming (including A) (inch pounds)	C Breakaway (inch pounds) (Minimum)
Stop and Retainer	30	300 ± 50	200
Head and Retainer	30	300 ± 50	200

3.3 Assembly.

3.3.1 Tensile load.- There shall be no movement or unlocking of the body from the thruster assembly when a tensile load of 30 to 40 pounds is applied.

3.3.2 Residual magnetism.- The thruster shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 The thruster shall not fail to fire.

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3.4.2 The thruster shall function without mechanical failure (See 6.1).

3.4.3 No load.- The thruster, fired with no load on the body, shall complete a stroke of 5.75 inches, minimum, and the body shall not separate from the tubes or retainer (see 6.2).

3.4.4 Load.- The thruster, fired under load, shall be mounted by the head threads at the free end of a 45 pound simulated canopy and at an angle normal to the simulated canopy. The thruster shall perform as follows:

3.4.4.1 Velocity.- The average velocity of the simulated canopy during the thruster stroke shall be 5 radians per second (14.6 feet per second), minimum.

3.4.4.2 Stroke.- Complete a stroke of 5.75 inches, minimum.

3.4.4.3 Rotation.- Rotate the 45 pound simulated canopy through an arc of 90 degrees.

3.4.4.4 Ignition delay.- Ignition delay shall not exceed 60 milliseconds (see 6.5).

3.4.5 Temperature.- The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65 to plus 200°F.

3.4.6 Initiation.- The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4 MS28741, @ 70° ± 5°F.

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components shall be free from cracks, splits, perforations and any similar defect.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with Specifications MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots, lot size and presentation of lots shall be in accordance with the "Submission of products", provisions of MIL-STD-105.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Minor		Major	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10,000	200	2	3	3	4

4.2.2.1 Cartridge, Thruster, M141, Dwg. D10521610, covering a detail of Dwg. F10521565.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual.

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
CRITICAL (Cont'd)	
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual

4.2.2.2 Head, Sub-Assembly consisting of Head and Pellet Assembly 10521583, Firing Pin 10521573, Shear Pin 10521574, Guide 10521572, O-Ring 501217 and Pellet 10520985, covering a detail of Dwg. F10521565.

CRITICAL	
1. Guide, improperly inserted	Manual & Visual
2. Any part missing	Visual
3. Excessive lubricant in head	Visual
4. Presence of shear pin in firing pin	Gage
MAJOR	
101. Threads damaged or not full	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.3 Retainer, Sub-Assembly, consisting of Retainer and Pellet Assembly 10521575, Inside Tube 10521578, Outside Tube 10521579, Stop and Pellet Assembly 10521582, O-Rings MS29513-16, 19, 22 and Cartridge 10521610, covering a detail of Dwg. F10521565.

CRITICAL	
1. Port in retainer, obstructed	Visual
2. Port in stop, obstructed	Visual
3. Cartridge missing	Visual
4. Improper assembly	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. O-ring damaged or missing	Manual & Visual
102. Threads damaged or not full	Visual
103. Protective finish inadequate or defective	Visual
104. Foreign matter	Visual
MINOR	
201. Pellet missing	Visual

4.2.2.4 Body and curved washer, sub-assembly consists of Body 10521568 and curved washer 10521580, covering a detail of Dwg. F10521565.

CRITICAL	
1. Curved washer missing or improperly installed	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Dwg. F10521565.

CRITICAL	
1. Any part missing as detected by x-ray	X-ray
2. Improper assembly	X-ray
3. Metal defective	Visual
MAJOR	
101. Total length (less shipping cap)	Gage
102. Threads damaged	Visual
103. Marking missing, incorrect or illegible	Visual
104. Varnish missing or inadequate	Visual

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4.2.2.5 (Cont'd).

Defect	Method of Inspection
MINOR	
201. Barrier material, missing	Visual
202. Protective finish inadequate or defective	Visual
203. Shipping cap missing	Visual
204. Foreign matter	Visual

4.2.2.6 Fiber Container, Dwg. C8800969.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Mental ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing Box, Dwg. C8800970.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual

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4.2.2.7 MINOR (Cont'd)

Defect	Method of Inspection
203. Strapping missing, loose, broken or improperly engaged	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each stop-retainer and head-retainer subassembly shall be subjected to a breakaway torque application. Failure of any thruster to comply with the requirement of 3.2 and Table I shall be cause for rejection of the thruster.

4.2.3.2 Residual magnetism.- Each thruster shall be subjected to the test for excessive residual magnetism. Failure of any thruster to comply with the requirement of 3.3.2 shall be cause for rejection of the thruster.

4.2.3.3 Tensile load.- Each thruster body shall be subjected to the tensile load application. Failure of any thruster to comply with the requirement of 3.3.1 shall be cause for rejection of the unit.

4.2.3.4 Functional tests.- The following tests shall be performed by the Government:

4.2.3.4.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.4.1 thru 3.4.6. The units shall be temperature conditioned as provided in Table III and 4.3.4.2. Failure of any unit to comply with the requirement or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

Table III

Sample Size	Method	Temperature
6	Load (See 3.4.4)	-65° to -70°F
6	Load (See 3.4.4)	$+65^{\circ}$ to $+75^{\circ}$F
6	Load (See 3.4.4)	$+160^{\circ}$ to $+165^{\circ}$F
1	No Load (See 3.4.3)	-65° to -70°F
1	No Load (See 3.4.3)	$+65^{\circ}$ to $+75^{\circ}$F
1	No Load (See 3.4.3)	$+160^{\circ}$ to $+165^{\circ}$F

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4.2.4 Inspection equipment.- The examination and tests shall be made using equipment in accordance with IEL-10521565.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, capable of measuring the torque requirements as specified in Table I shall be used with an appropriate adapter to determine the adequacy of the joints. Both joints may be checked in one operation.

4.3.2 Tensile load.- Upon completion of assembly, a tensile load of 30 to 40 pounds shall be applied to the body of the thruster to determine that the shear screw is properly engaged.

4.3.3 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

4.3.4 Functional tests.

4.3.4.1 Ballistics.- The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.3.4.2 Temperature conditioning time.- The thrusters for ballistic tests shall be temperature conditioned for a minimum of 3 hours.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A - Packing and marking shall be in accordance with Drawings C8800969 and C8800970.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES.

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6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 No load.- Permanent yielding of the body or tubes shall not be cause for rejection when the thrusters are tested with no load on the body.

6.3 Load.- The expected performance of the thruster under conditions simulating those existing in aircraft is obtained when thrusters satisfactorily meet conditions of the load firing.

6.4 "Ignition delay".- is defined as the time lapse between the firing of the cartridge and the beginning of the continuous pressure rise in the body of the thruster.

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Preparing activity:

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